

MICHAEL POTTS

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GitHub: <https://github.com/michael-j-potts/Portfolio>

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To whom it may concern,

What makes data truly valuable? In my experience, it's not just the ability to analyze it, but the ability to turn it into decisions that meaningfully improve outcomes for real people. That's the kind of work that motivates me and the reason I'm excited about opportunities in applied data science and machine learning.

In my current role with the Alberta Native Friendship Centres Association, I build and maintain Python pipelines to extract, clean, and analyze large amounts of structured and unstructured data from multiple sources. These tools support operational planning and program evaluation across 21 Friendship Centres in Alberta. I also regularly present findings at conferences and breakout sessions to groups of 40–50 people. This has made me very comfortable with public speaking on complex topics. I am most proud of my work on this project as it utilizes data and the analysis process to impact the lives of thousands of Indigenous children and youth across the province of Alberta.

I hold a Master of Computer Science with a specialization in Data Science, where I built a strong foundation in machine learning, statistics, and data analysis. During my internship at Sunstella, I developed machine learning models using graph neural networks and neural-backed decision forests for patient trajectory modeling with ICD-10 data. This work involved dataset preparation, feature engineering, experimentation, and building reproducible workflows using Python and Pytorch.

Previously, at Canuck Place Children's Hospice, I worked closely with clinicians and interdisciplinary teams to analyze healthcare data and identify trends in pediatric populations. I applied statistical modeling and exploratory data analysis techniques, while communicating findings through reports, presentations, and visualizations designed for non-technical audiences. This experience strengthened my ability to work with cross-functional teams and turn complex data into practical insights.

I'm proficient in Python and experienced with machine learning frameworks including Pytorch and scikit-learn, along with visualization tools such as Matplotlib and Seaborn. I'm comfortable working in Linux environments, using Git, and collaborating on technical projects. I'm motivated to continue growing my skills in machine learning, applied analytics, and large-scale data systems.

Sincerely,

Michael Potts

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Passionate data professional, with 6 years of experience turning complex data into actionable insights. Proficient in Python, R, C++, and SQL, with hands-on experience in machine learning, data analysis, and visualization. I'm excited to bring my skills in Data Science and Machine Learning to a dynamic team where I can help drive meaningful results through data-driven strategies.

EXPERIENCE

DATA SCIENTIST

Alberta Native Friendship Centres Association (ANFCA)

MARCH 2024 - PRESENT

Worked with the education team on a \$7.5 million grant from Alberta Education to evaluate academic support programs for Indigenous children and youth. I transformed thousands of records from Word into structured Excel datasets for large-scale statistical analysis in Python. In order to do this, I manually programmed several extraction and analysis scripts in python. I successfully automated around 80-90% of the extraction and analysis process to improve turnaround time. Further, I presented my findings 10+ times to audiences of 40-50 people on average at various conferences.

STATISTICIAN AND HEALTH RESEARCHER

Canuck Place Children's Hospice

MARCH 2023 – JANUARY 2024 (MAT LEAVE CONTRACT)

I collaborated with the medical director on palliative care research, applying advanced statistical methods to analyze health data. I designed and executed machine learning models, geo-analytical patient and doctor mappings, and predictive analytics to forecast palliative care needs across British Columbia. Each of these were programmed primarily in python. Finally, I produced data-driven insights to inform healthcare planning and resource allocation for the palliative care team.

DEEP LEARNING RESEARCHER

Sunstella Machine Learning Researcher Student Program

MAY 2022 – JULY 2022

I worked on a pediatric classification program utilizing graph neural networks and neural backed decision forests to classify the severity and level of care required for respiratory infections.

RESEARCH ASSISTANT

TLEC Innovations

JANUARY 2020 – SEPTEMBER 2021

Conducted research on chemical catalyst structures to support the development of an exhaust recapture prototype. I assisted with experimental design, data collection, and analysis to help the lead researcher. Further, I collaborated with the senior researcher on interpreting results and preparing technical reports and presentations for investors.

EDUCATION

MASTERS OF COMPUTER SCIENCE

University of Illinois – Urbana Champaign

Achieved December 2022 – GPA 4.0

Relevant Coursework: Applied Machine Learning, Deep Learning for Healthcare, Advanced Bayesian Modeling, Data Visualization

BACHELOR OF SCIENCE IN NURSING

University of British Columbia - Okanagan

Achieved June 2017 – GPA 3.95

Specializations: Research, Cardiac Nursing, Community Wound Care

RECENT SPEAKING CONFERENCES

At each of the following conferences, I presented on the educational programs data and data analyses. At each conference, I presented in front of 40 – 50 people on average in conference breakout rooms.

College of Alberta School Superintendents - APRIL 2026

Ever Active Schools: Shaping the Future - JANUARY 2026

Gender Based Sexual Violence Knowledge Exchange: ANFCA - JUNE 2025

Education Knowledge Exchange: ANFCA - APRIL 2025

Alberta Teachers Association: Endless Skies - FEBRUARY 2025

Ever Active Schools: Shaping the Future - JANUARY 2025

Child Bright: Pain and Symptoms in Children with Neurological Impairment – JUNE 2023

Canuck Place Children's Hospice: AI and its' place in health care – MAY 2023

BOARDS

I sat on each of the boards listed as a representative of **ANFCA** contributing to strategic planning and community engagement.

Indigenous Education Committee - SEPTEMBER 2025 – PRESENT

Naloxone Steering Committee - DECEMBER 2024 – PRESENT

CERTIFICATES

Indigenous Canada - MARCH 2024

Microsoft Learning with TensorFlow on Google Cloud – FEBRUARY 2024

Microsoft Azure Data Science Associate – JANUARY 2024

Data or Specimens Only Research: MIT – OCTOBER 2022

PROJECTS

Deep Learning – Assignments covering deep learning, CNNs, Seq2Seq, RNNs, autoencoders, attention models, GNNs, memory networks and generative models. Recreated and improved upon a latent space correlational neural network model.

IOT – Created a diabetic foot monitor and app that monitors feet for trauma and proper care. Created a self-driving car that accepts coordinates, implements object detection and avoidance as well as intelligently handles different types of obstacles.

Applied Machine Learning – Assignments covering classification, regression, clustering, expectation maximization, high dimensional data, and convolutional neural networks. Project predicting vehicle GHG emissions on 16.3 million vehicle data set.

SKILLS

Programming Languages and Development: Python, SQL, C++, R, Git, Github, Rest APIs, JSON, Linux

Tools and Technologies: Excel, Microsoft Azure, AWS, Databricks, RedCap, Docker, ETL Pipelines

Data Analysis: Data Cleaning (OpenRefine), SPSS, Apache Spark

Machine Learning: Classification, Time Series Analysis, Regression, Clustering, NLP, PyTorch, Tensorflow, Sci-kit Learn

Data Visualization: Tableau, PowerBI, Matplotlib, Seaborn, Plotly